Course Schedule

Schedule is subject to change. All changes are posted here. If a due date here conflicts with Canvas, this page is correct.

Materials link up as they are released; a week with no links is not published yet.

Module 1: Foundations

Why do we care about quantitative representations of climate hazard?

Week	Date	In class	Due
1	Mon 8/24	Welcome to CEVE 543 · Course logistics and expectations	
	Wed 8/26	Return periods and exceedance probabilities · Return periods in Julia	Week 01 Practice Problems (Optional); Week 01 Reading: Motivations (nothing to submit)
	Fri 8/28	Monte Carlo simulation · Monte Carlo in Julia · Lab 1: Monte Carlo convergence	Lab 1 software install
2	Mon 8/31	Intro to the climate system · Climate as a dynamical system	Week 02 Practice Problems (Optional); Week 02 Reading: Climate as a dynamical system
	Wed 9/2	Autocorrelation, partial autocorrelation, and the spectrum	
	Fri 9/4	Lab 2: Climate as a dynamical system	Lab 1 due
3	Mon 9/7	*No class (Labor Day)*	
	Wed 9/9	Estimating parameters from data · Estimating parameters from data	Week 03 Practice Problems (Optional); Week 03 Reading: Inference and resampling (nothing to submit)
	Fri 9/11	Bootstrap · Lab 3: Estimation and uncertainty	Lab 2 due

Module 2: Univariate extremes

How do we characterize the probability distribution of a single hydroclimate hazard at a single location, and what are the associated uncertainties?

Week	Date	In class	Due
4	Mon 9/14	Week 4 overview · The extremal types theorem and the GEV	Week 04 Practice Problems (Optional); Week 04 Reading: Extreme value theory
	Wed 9/16	Fitting a GEV in Julia	
	Fri 9/18	Lab 4: Fitting a GEV to a tide gauge	Lab 3 due
5	Mon 9/21	Week 5 overview · The GPD and compound-Poisson model · Peaks over threshold in practice	Week 05 Reading: Peaks over threshold

Week	Date	In class	Due
	Wed 9/23	Poisson counts vs exponential waiting times	Week 05 Practice Problems (Optional)
	Fri 9/25	Lab 5: Peaks over threshold in practice	Lab 4 due
6	Mon 9/28	Week 6 overview · Trend tests · Nonstationary models in practice · Example: nonstationarity and the 100-year flood	Week 06 Practice Problems (Optional); Week 06 Reading: Nonstationary flood frequency at Brays Bayou (bring two questions Wednesday)
	Wed 9/30	Case study: Brays Bayou flood frequency	
	Fri 10/2	Lab 6: Is extreme rainfall at Hobby changing?	Lab 5 due
7	Mon 10/5	Week 7 overview · Regionalization · Bayesian pooling	
	Wed 10/7		Lab 6 results to present in class
	Fri 10/9	*No in-person meeting (asynchronous, instructor at the ASA ENVR workshop)* · Lab 7: Regional frequency analysis with a CO2 trend	Lab 6 due

Module 3: Spatiotemporal extremes

How do we sample hydroclimate hazard(s) across space and/or time to characterize complex extremes?

Week	Date	In class	Due
8	Mon 10/12	*No class (Midterm recess)*	
	Wed 10/14		Practice Problems TBD; Reading TBD
	Fri 10/16	Test 1	Lab 7 due
9	Mon 10/19	Compound events	Practice Problems TBD; Reading TBD
	Wed 10/21	Joint and conditional distributions · Copulas: figures	
	Fri 10/23	Project work session	
10	Mon 10/26	Week 10 overview · Storm transposition	Selection due; Practice Problems TBD; Reading TBD
	Wed 10/28	The transposition integral in practice	
	Fri 10/30	Project work session	
11	Mon 11/2	Week 11 overview · Climate models, downscaling, and bias correction	Practice Problems TBD; Reading TBD
	Wed 11/4	Quantile mapping and delta methods	
	Fri 11/6	Lab 8: Downscaling rainfall with quantile mapping and weather types	
12	Mon 11/9	Week 12 overview · PCA and EOF analysis · Weather typing	Practice Problems TBD; Reading TBD
	Wed 11/11	PCA and k-means on reanalysis fields	
	Fri 11/13	Lab 8, continued	

Week	Date	In class	Due
13	Mon 11/16	Week 13 overview · Markov chains and hidden Markov models · Climate-informed generators	Practice Problems TBD; Reading TBD
	Wed 11/18	Building a synthetic generator	Project checkpoint
	Fri 11/20	Project work session	Lab 8 due (two-week lab)
14	Mon 11/23	Peer review of project methods	
	Wed 11/25	*No class (Thanksgiving)*	
	Fri 11/27	*No class (Thanksgiving)*	

Synthesis

Week	Date	In class	Due
15	Mon 11/30	Test 2	
	Wed 12/2	Discussion	
	Fri 12/4	Synthesis and wrap-up	Paper, memo, code, and AI use statement due

Bibliography